Executive Summary & Core Positioning

1. The Operational Problem: Coarse Grids Miss Coastal Physics

Global physical products—such as those from ECMWF and Copernicus Marine Environment Monitoring Service (CMEMS)—provide essential ocean-state baselines at $4.2\ \text{km}$ to $25\ \text{km}$ horizontal resolutions. However, coarse global grids smooth out critical coastal bathymetric features, flatten narrow island channels (e.g., Freus channel between Ibiza and Formentera, Dragonera channel off Mallorca), and underestimate wave-current interactions and local wind-sea generation.

2. The Commercial & Coastal Value of 1 km Resolution

PredSea bridges this operational gap with an explicit positioning philosophy:

> **Ocean data is everywhere. Operational decisions are not.**

By downscaling regional ocean state data onto a high-resolution $\mathbf{1\ km}$ **curvilinear grid** (401×501 nodes across 32 vertical σ-layers in the Balearic Basin), PredSea provides: * **Sub-Kilometer Coastal Accuracy:** Resolves steep shoreline bathymetry, narrow island channels, and localized coastal upwelling anomalies. * **Predictive Decision Intelligence:** Converts raw 3D velocity vectors and wave spectra into vessel response metrics, port safety alerts, and captain-facing advice. * **Unmatched Unit Economics:** Delivers regional scale high-resolution forecasts at ~$2.22/day using serverless GCP Batch Spot execution—a >90% reduction compared to legacy HPC infrastructure.

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|                                 PredSea Platform                                  |
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|  +--------------------+    +--------------------+    +-------------------------+  |
|  |   WRF Atmosphere   |    |    SWAN Waves      |    |       CROCO Ocean       |  |
|  |  (10m Wind, Flux)  |    | (Hs, Tp, Spectrum) |    |  (3D u, v, T, S, Zeta)  |  |
|  +---------+----------+    +---------+----------+    +------------+------------+  |
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|                  +---------------------------------------+                        |
|                  |      GCP Batch Spot Execution         |                        |
|                  +-------------------+-------------------+                        |
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|                  +---------------------------------------+                        |
|                  |    Canonical NetCDF4 + GCS Pipeline   |                        |
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|                  | BigQuery Decision Engine & REST APIs  |                        |
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Key Performance Indicators (KPIs)

Parameter	Legacy Cluster Model	PredSea Cloud-Native Architecture
Execution Paradigm	Fixed On-Prem / Monolithic VM	**Serverless GCP Batch Spot Instances**
Grid Resolution	$4.2 \text{ km} - 10 \text{ km}$ (Global/Regional)	$\mathbf{1.0 \text{ km}}$ **(High-Resolution Coastal Tiles)**
Pre-Processing Time	40 minutes (Single-threaded)	**1 min 42 sec** (Python `ProcessPoolExecutor`)
Total Western Med Compute Time	> 28 Hours (Sequential)	**< 35 Minutes** (Parallel Region Spot Execution)
Daily Infrastructure Cost	~$45.00 / day	**~$2.22 / day** (Spot VM + GCS + BigQuery)
Validation Benchmark	$H_s \text{ RMSE} \approx 0.42 \text{ m}$	$\boldsymbol{H_s} \textbf{ RMSE} \leq \mathbf{0.11 \text{ m}}$ (SOCIB Buoy Ground Truth)

Table of Contents

The complete technical specification is structured across six dedicated sub-documents: